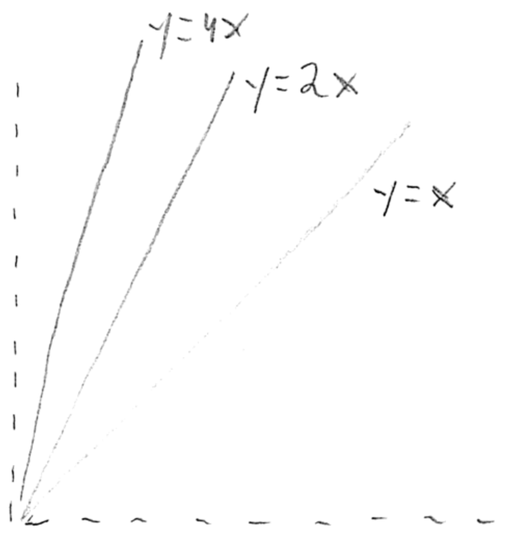
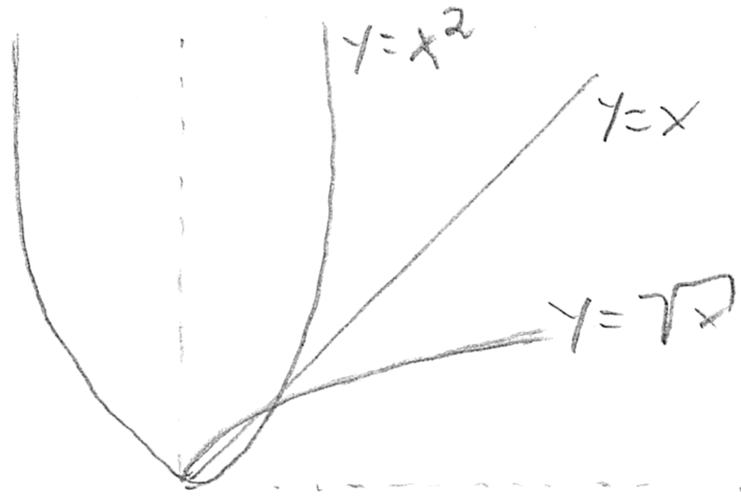
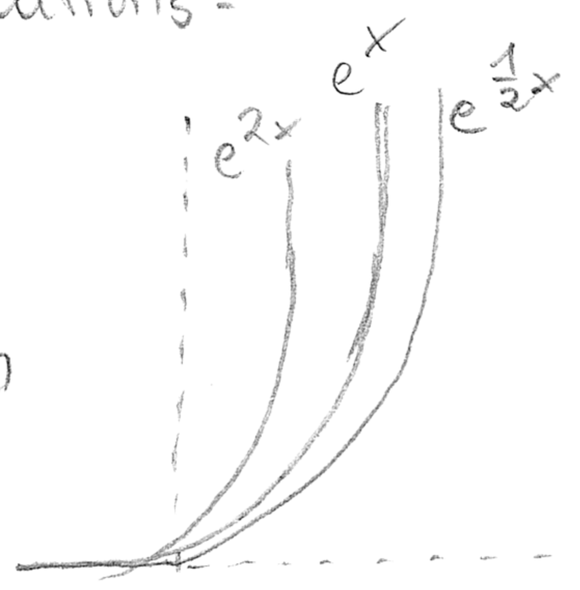
NMRC Math Exercises - Solutions -

1)



a)





b)

2) $\boxed{f(x) = 3x^x}$ (4)

- Note that our variable is in the base <u>and</u> in the exponent!
- Thus what would make our life easier is "getting rid" of x in either the base or the exponent. (Then we can apply, e.g. the chain rule)

↳ For this, we can exploit one of the logarithmic rules/identities:

$$x^x = e^{x \cdot \ln(x)}$$

→ So we re-write:

$$f(x) = \underbrace{3 \cdot e^{x \cdot \ln(x)}}_{v}$$, and now use chain rule:

- Outer function $v = 3e^u \rightarrow v' = 3e^u$
- Inner function $u = x \cdot \ln(x) \rightarrow u' = x \cdot \frac{1}{x} + 1 \cdot \ln(x)$

$$= 1 + \ln(x)$$

[Product Rule applied]

→ Now we have both parts, so we can put them together (according to chain rule): $f(x) = v(u(x))$

↳ $f'(x) = f'(v(x) \cdot u'(x)$, so:

$$f'(x) = \underbrace{3 \cdot e^{x \cdot \ln(x)}}_{= f'(v(x))} \cdot \underbrace{(1 + \ln(x))}_{= u'(x)}$$

- Now we can simplify, using log identity again:

$$= 3x^x \cdot (\ln(x) + 1)$$

- which we could re-write, if we wanted:

$$= 3x^x \cdot \ln(x) + 3x^x$$

$$= 3 \cdot (x^x \cdot \ln(x) + x^x)$$

2) $\boxed{f(z) = 2e^z + \frac{1}{z^2}}$ (5)

- We see a sum with two components/terms, so we take the derivative of each part separately:

→ $f'(2e^z) = \underline{2e^z}$
→ $f'(\frac{1}{z^2})$ → we rewrite $\frac{1}{z^2}$ as z^{-2}, so we can build:
$= f'(z^{-2}) = \underline{-2z^{-3}}$ (using the power rule)

- Now we put them together again:
$f'(z) = 2e^z - 2z^{-3}$

- Which we could re-write as:
$f'(z) = 2e^z - \frac{2}{z^3}$

2) $\frac{d}{d\varphi}(e^{4\varphi^2})$

- Nested function, so chain rule:

→ outer function: $f(u) = e^u$ (where $u = 4\varphi^2$)
→ Inner function: $g(x) = 4\varphi^2$

→ Derivative of outer function is just e^u, so $e^{4\varphi^2}$
→ Derivative of inner function is 8φ (power rule)

- Put everything together (chain rule):

$\frac{d}{d\varphi}(e^{4\varphi^2}) = e^{4\varphi^2} \cdot 8\varphi$
$= 8\varphi e^{4\varphi^2}$

2) $\boxed{\frac{d}{dx}(\cos^2(x))}$ (7)

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- Rewrite $\cos^2(t)$ as $(\cos(t))^2$
- Then we consider the function as $f(t) = (g(t))^2$, where $g(t) = \cos(t)$
  (because $\cos(t)$ is nothing else than a function)

- So we have a nested function in front of us and can use the chain rule

↳ If $f(t) = (g(t))^2$, then
$f'(t) = 2 \cdot g(t) \cdot g'(t)$

↳ So in this case we get
$g(t) = \cos(t)$, so
$g'(t) = -\sin(t)$

- Now we can use the chain rule and plug in $g(t)$ and $g'(t)$:

$$\frac{d}{dx}(\cos^2(t)) = 2 \cdot \cos(t) \cdot (-\sin(t))$$

$$= -2\cos(t) \cdot \sin(t)$$

*
we could simplify this further, using a trigonometric identity:
$[\sin(2x) = 2\sin(x) \cdot \cos(x)]$

↳ thus:

$$-2\cos(t) \cdot \sin(x) = -\sin(2x)$$

2) $\frac{d}{dx}\left(\frac{x}{1+e^{x^2}}\right)$ (8)

4/10

- We see a quotient → use quotient rule,
- And chain rule for e^{x^2}

↳ Quotient Rule: If $f(x) = \frac{u}{v}$, then $f'(x) = \frac{u' \cdot v - u \cdot v'}{v^2}$

↳ our $\underline{u \text{ is } x}$, so $\underline{u' = 1}$
our $\underline{v \text{ is } 1+e^{x^2}}$ (so chain rule ...)
so $v' = 0 + 2x \cdot e^{x^2}$

- Now we put together everything using the quotient rule:

$$\frac{d}{dx} = \frac{1 \cdot (1+e^{x^2}) - (x \cdot (2x \cdot e^{x^2}))}{(1+e^{x^2})^2}$$

$$= \frac{1 + e^{x^2} - 2x^2 e^{x^2}}{(1+e^{x^2})^2}$$

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3) $S(x) = \frac{1}{1+e^{-x}}$

- Quotient, so → quotient rule: $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u' \cdot v - u \cdot v'}{v^2}$

↳ in our case: $u = 1$
$v = 1 + e^{-x}$

So: $u' = 0$ (constant Rule)
$v' = 0 - e^{-x}$ (chain Rule)

- So we plug that into our quotient rule expression:

$$S'(x) = \frac{0 \cdot (1+e^{-x}) - 1 \cdot (-e^{-x})}{(1+e^{-x})^2}$$

$$S'(x) = \frac{e^{-x}}{(1+e^{-x})^2}$$

- Now let's take a look again at our initial expression: $S(x) = \frac{1}{1+e^{-x}}$
- We could easily re-write this as:
- And recognize the similarity → $1 + e^{-x} = \frac{1}{S(x)}$
(same term, but squared in the above expression)

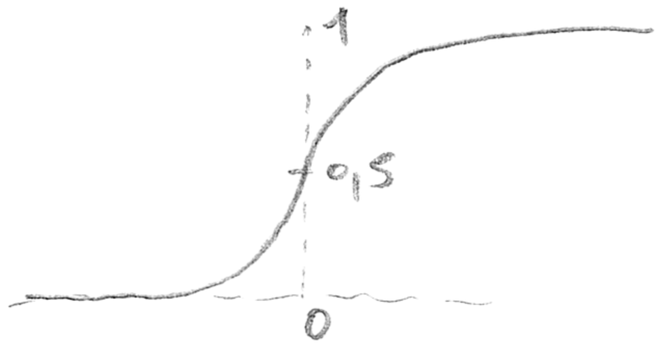
- So let's square the expression here on both sides, so we get:
- $(1+e^{-x})^2 = \left(\frac{1}{S(x)}\right)^2$
- Compare that to • $S'(x) = \frac{e^{-x}}{(1+e^{-x})^2}$ and note that we see the same expression!

- This allows us to re-write $S'(x) = \frac{e}{(1+e^{-x})^2}$ as: $S'(x) = \frac{e}{\left(\frac{1}{S(x)}\right)^2}$
- Now we use a similar trick again, and:
express e^{-x} in terms of our original sigmoid function:
  - If $S(x) = \frac{1}{1+e^{-x}}$, then $e^{-x} = \frac{1-S(x)}{S(x)}$
- Plug that into our equation:

$$S'(x) = \frac{1-S(x)}{S(x)} \cdot S(x)^2$$ | Divide by $S(x)$, we get

$$= (1 - S(x)) \cdot S(x)$$

4) Drawing-part:



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4) we have $f(x) = \frac{1}{1+e^{-kx}}$ with $k=2$

- If k increases, the term e^{-kx} becomes smaller
  ↳ so if x is positive, then: $\frac{1}{1+e^{-kx}}$ gets closer to 1 (= increases), because we more or less approach $\frac{1}{1}$

  ↳ if x is negative, then the term $\frac{1}{1+e^{-kx}}$ gets closer to zero (because $1+e^{-kx}$ gets larger)

- Thus, if k increases, we get a steeper discrimination!

5) $\frac{d}{dx}(\tanh(x)) = \frac{d}{dx}\left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right)$

7/10

- Quotient, so quotient rule: If $z = \frac{u}{v}$, then $z' = \frac{u' \cdot v - u \cdot v'}{v^2}$
- In our case:

$u = e^x - e^{-x}$
$v = e^x + e^{-x}$ and we differentiate each part seperately (meaning e^x and e^{-x}, separately)

- The derivative of $e^x = e^x$,
- But for e^{-x}, we need the chain rule (which is: If $z = f(g(x))$, then $z' = f'(g(x)) \cdot g'(x)$

↳ So we treat e^{-x} as outer function, and $-x$ as inner function:

$f(e^x) \rightarrow f'(e^x) = e^x$
$g(-x) \rightarrow g'(-x) = -1$

↳ And we get: the derivative of e^{-x} is $e^{-x} \cdot (-1) = -e^{-x}$

- Now we have everything to put all terms in our quotient rule term:

$u = e^x - e^{-x} \rightarrow u' = e^x + e^{-x}$
$v = e^x + e^{-x} \rightarrow v' = e^x - e^{-x}$

↳ So:

$$\frac{d}{d(x)}\left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right) = \frac{(e^x + e^{-x}) \cdot (e^x + e^{-x}) - (e^x - e^{-x}) \cdot (e^x - e^{-x})}{(e^x + e^{-x})^2}$$

$$= \frac{(e^x + e^{-x})^2}{(e^x + e^{-x})^2} - \frac{(e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

$$= 1 - \frac{(e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

Recognize that this term is nearly identical to our original tanh function (only squared)

So, $= 1 - \tanh^2(x)$

6) $f(x,y) = x^3 + 3xy + 4y^2 + x^2y$ (9)

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- $\frac{\partial f}{\partial x} = 3x^2 + 3y + 2xy$
- $\frac{\partial f}{\partial y} = 3x + 8y + x^2$

$f(x,y,z) = x^2 + 3xyz$ (10)

$\frac{\partial f}{\partial x} = 2x + 3yz$

$\frac{\partial f}{\partial y} = 3xz$

$\frac{\partial f}{\partial z} = 3xy$

$f(x,y) = x^y$ (11)

$\frac{\partial f}{\partial x} = y \cdot x^{y-1}$

$\frac{\partial f}{\partial y}$ is a bit more tricky, because the variable is in the exponent

- We need to know/remember that $\frac{d}{dx}(a^x) = a^x \cdot \ln(a)$, so:

$\frac{\partial f}{\partial y} = x^y \cdot \ln(x)$

7) $\boxed{f(x,y) = x^2 y}$ (13)

$$\nabla f(x,y) = \begin{pmatrix} 2xy \\ x^2 \end{pmatrix}$$

* Remember:
The gradient is a vector which contains the partial derivatives!

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$$\boxed{f(x,y,z) = x^2 y \cos(z)} \quad (14)$$

$$\nabla f(x,y,z) = \begin{pmatrix} 2xy\cos(z) \\ x^2 \cos(z) \\ -x^2 y \sin(z) \end{pmatrix}$$

$$\boxed{f(x,y,z,t) = x^2 y t^{-1} \cos(z)} \quad (15)$$

$$\nabla f(x,y,z,t) = \begin{pmatrix} 2xyt^{-1} \cdot \cos(z) \\ x^2 t^{-1} \cdot \cos(z) \\ -x^2 y t^{-1} \cdot \sin(z) \\ -x^2 y t^{-2} \cdot \cos(z) \end{pmatrix}$$

Note: It is tempting to think that all variables which are kept constant should disappear because the derivative of a constant is zero.

However, the variables in 13, 14, and 15 appear as a product!

[Multiple constant Rule: $\frac{d}{dx}(c \cdot f(x)) = c \cdot f'(x)$] !

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8) $v = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$, $u = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$

$v \cdot u = (3 \cdot 1) + (2 \cdot 2) + (1 \cdot 4)$

$= 3 + 4 + 4$

$= 11$

9) $A + B = \begin{bmatrix} 1+0 & 2+1 & 3+1 \\ 4-1 & 5+0 & 6+1 \\ 7-1 & 8-1 & 9+0 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 4 \\ 3 & 5 & 7 \\ 6 & 7 & 9 \end{bmatrix}$ (19)

$A + 2B = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 5 & 8 \\ 5 & 6 & 9 \end{bmatrix}$ (20)

$A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$, $A^T - 2B = \begin{bmatrix} 1 & 2 & 5 \\ 4 & 5 & 6 \\ 5 & 8 & 9 \end{bmatrix}$ (21)